Q.24 With the help of suitable block diagram, explain the working of delta modulation (DM) system.

Q.25 Write a short note on PPM.

Q.26 Define amplitude shift keying.

Q.27 What do mean by PSK?

Q.28 What are digital modulation and digital modulation techniques?

Q.29 Discuss noise and distortion.

Q.30 Write note on: (a) Transmission mode (b) Equalizer

Q.31 Differentiate between synchronous and asynchronous transmission.

Q.32 List various features of modem.

Q.33 Write short note on Modem interconnections.

Q.34 Explain various modes of Modem operations.

Q.35 Explain any two switching technique.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain the basic block diagram of digital and data communication system.

Q37 Describe the principle of pulse code modulation (PCM) along with block diagram.

Q.38 Differentiate between ASK, FSK and PSK. Which one is best among them?

Roll No. ....

5th Sem / Eltx.

Sub : Digital Communication

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The modulation techniques used to convert analog signal into digital signal are

a) Pulse code modulation

b) Delta Modulation

c) Adaptive delta Modulation

d) All

Q.2 The factors that cause quantizing error in delta modulation are:

a) Slope overload distortion

b) Granular Noise

c) A & B both

d) None

Q.3 Digital signals:

a) do not provide a continuous set of values

b) Can utilize decimal of binary systems

c) Represent values as discrete steps

d) All of above

Q.4 The main advantage of PCM system is
a) Lower noise
b) Lower power
c) Lower bandwidth
d) None of these

Q.5 Types of Modulation used in MODEMs.
a) FSK b) QPSK
c) QAM d) All above

Q.6 Blocking is not a problem in a _____ switch.
a) Single stage b) Two stage
c) Multistage d) Three stage

Q.7 A repeater is used to:
a) Reduce noise
b) Amplify and regenerate signals
c) Encode data
d) Compress data

Q.8 Bandwidth of a signal represents:
a) Signal strength
b) Range of frequencies
c) Transmission speed only
d) Noise level

Q.9 In differential Pulse Code modulation techniques the decoding is performed by
a) Sampler b) Accumulator
c) PLL d) Quantizer

Q.10 Advantages of digital communication are:
a) Easy multiplexing b) Easy processing
c) Reliable d) All of mentioned

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define PPM

Q.12 Define DPCM.

Q.13 Define FSK.

Q.14 Define QPSK.

Q.15 Define Data Communication.

Q.16 Define PWM

Q.17 Define FREQUENCY HOPPING.

Q.18 Define BANDWIDTH

Q.19 Define SAMPLING.

Q.20 Define USART.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 What are the various modes of data communication? Name them.

Q.22 What are the advantages and disadvantages of digital communication system?

Q.23 What do you mean by Quantization and how it is performed?